



Department of Computer Technology

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027(ODD)

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	2026-27 (ODD)	Course Name	LAB: Software Engineering
Semester	7th	Course Code	23CT1704
Roll No	25	Name of Student	Tanushri Sawalikar

Practical Number	01
Aim	Introduction to Software Engineering fundamentals, UML and StarUML
Theory (100 words)	1. Introduction to SE (Software Engineering) Software Engineering is the systematic, disciplined, and quantifiable approach to the development, operation, and maintenance of software. It applies engineering principles to create software that is reliable, efficient, scalable, and economical. 2. SDLC (Software Development Life Cycle) The SDLC is a structured process used by IT industries to design, develop, and test high-quality software. 1. Requirement Analysis: Gathering business needs from stakeholders to define the project's exact scope. 2. System Design: Creating the architectural blueprint, database design, and interface mockups based on requirements. 3. Implementation: Writing the actual source code using appropriate programming languages and frameworks. 4. Testing: Verifying that the code works correctly, fixing bugs, and ensuring it meets user expectations. 3. What is UML? (Definition & Advantages) Unified Modeling Language (UML) is a standardized, general-purpose modeling language used to visualize, specify, construct, and document the artifacts of a software system. • Advantages: - o Standardization: Provides a universal language that developers, architects, and business analysts all understand. - o Clarity: Simplifies complex system design by breaking it down into visual blueprints.



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- o Better Planning: Helps identify design flaws, system bottlenecks, and dependencies before writing code.
- o Documentation: Acts as a permanent visual reference for future maintenance and scaling.

4. Types of UML Diagrams

UML diagrams are divided into two main categories: static structures and dynamic behaviors.

Structural Diagrams (Static Views)

- Class Diagram: Shows the system's classes, attributes, operations, and the static relationships between them.
- Object Diagram: Captures a snapshot of the instances (objects) of classes at a specific point in time.
- Component Diagram: Illustrates how software components (libraries, executables, modules) are organized and wired together.
- Deployment Diagram: Models the physical hardware runtime environment, showing where software components reside.
- Package Diagram: Organizes system elements into groups or namespaces to show structural dependencies.

Behavioral Diagrams (Dynamic Views)

- Use Case Diagram: Models how external actors interact with the system to achieve specific functional goals.
- Sequence Diagram: Shows how objects interact in a sequential time order to execute a specific workflow.
- Activity Diagram: Represents the step-by-step procedural flow of control or data within a system (like a flowchart).
- State Diagram: Models the lifecycle of a single object as it changes states in response to external events.
- Communication Diagram: Focuses on the structural organization of objects that send and receive messages.

5. Important UML Diagrams & Components

Your lab specifically highlights three crucial diagrams required for practical software design:

A. Use Case Diagram

Maps out the system functions from a user's perspective.

- Actor: An external entity (human, device, or external system) that interacts with the system. Represented by a stick figure.
- Use Case: A specific function or task performed by the system. Represented by an oval shape.
- System Boundary: A box drawn around the use cases to define what is inside

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the
system versus what is external.

- Example:

- o Actor: Customer

- o Use Case: Login, View Product, Make Payment

B. Class Diagram

The foundation of object-oriented design, representing the structural building blocks of your application.

- Each class is represented by a box split into three horizontal sections:

1. Top: Class Name (e.g., User)

2. Middle: Attributes/Variables(e.g., - email: String)

3. Bottom: Operations/Methods (e.g., + verifyPassword())

C. Sequence Diagram

An interaction diagram detailing how operations are carried out over time.

- Lifelines: Vertical dashed lines representing the lifespan of an object during the interaction.

- Activation Boxes: Thin vertical rectangles on the lifelines showing when an object is actively executing a task.

- Messages: Horizontal arrows pointing from sender to receiver showing function calls and returns.

6. UML Relationships

Relationships define how different elements in your design connect and interact with one another.

- Association: A structural relationship where one object uses or interacts with another object. (e.g., A Driver drives a Car). Represented by a straight solid line.

- Aggregation: A part-whole relationship where the child can exist independently of the parent. (e.g., A Department has Professors. If the department closes, the professors still exist). Represented by a clear diamond arrow.

- Composition: A strict part-whole relationship where the child cannot exist without the parent. (e.g., A House has Rooms. If the house is destroyed, the rooms cease to exist). Represented by a solid black diamond arrow.

- Inheritance / Generalization: A relationship where a child class inherits attributes and methods from a parent class. (e.g., A Dog is an Animal). Represented by a solid line with a hollow triangle arrow.

- Dependency: A weaker relationship where a change in one class may affect another class that relies on it. Represented by a dashed arrow.

7. UML Symbols:

UML Element	Symbol / Visual	Main Purpose
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Class	Rectangle	Defines object structure and behavior
Actor	Stick figure	Represents a user or external system
Use Case	Oval	Represents a system goal or function
Composition	Filled Diamond	Strong ownership (parts die with whole)
Aggregation	Hallow Diamond	Weak ownership (parts survive alone)
Relationship	Lines/Arrows	Connects items to show how they interact
Decision	Diamond Shape	Shows a branch or choice in a workflow

8. Introduction to Rational Rose**Definition**

Rational Rose is an automated, component-based Visual Modeling tool designed to build, document, and manage object-oriented software architectures using the Unified Modeling

Language (UML). It acts as a graphical design platform that converts abstract business logic into structural software models, directly linking the design phase with source code generation.

Applications

- System Architecture Design: Mapping out complex software structures, databases, and system topologies before development starts.
- Legacy Code Modernization: Reverse-engineering older, undocumented codebases into visual models to understand how they work.
- Enterprise Business Modeling: Documenting organizational workflows, user permissions, and business processes using Use Case and Activity diagrams.
- Cross-Language Code Skeleton Generation: Creating base structural frameworks for multi-language projects (e.g., generating C++, Java, or Ada files from a single model).

Advantages

- Round-Trip Engineering: Keeps diagrams and source code synchronized automatically, reducing manual documentation updates.



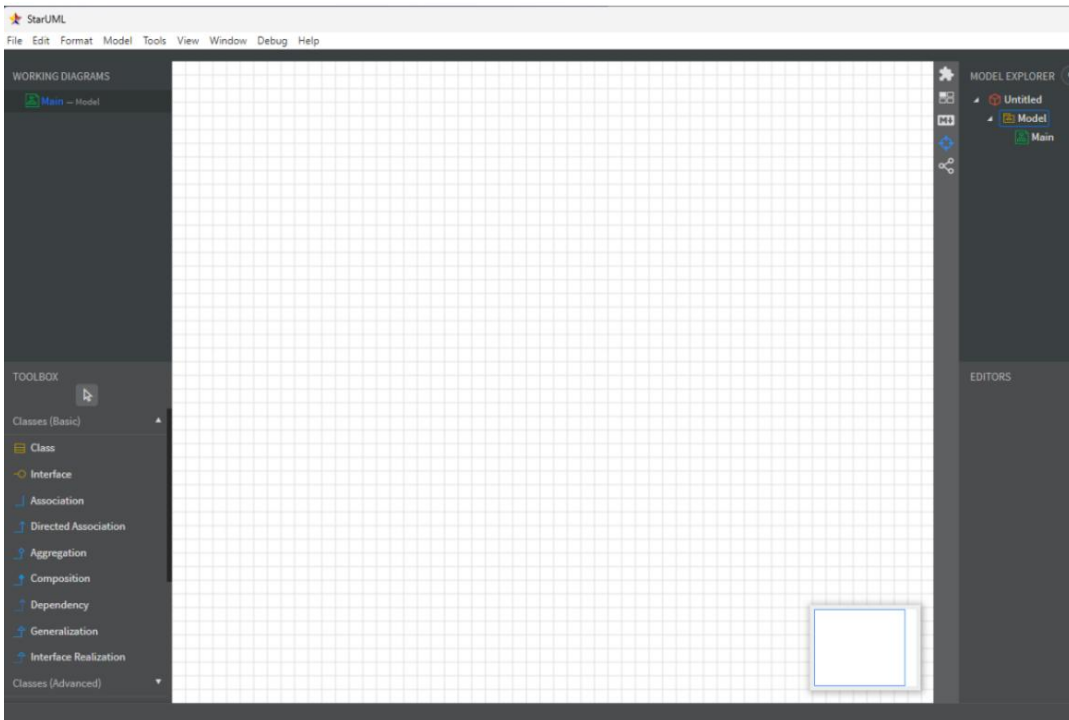
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	• Error Prevention: Detects architectural flaws, logic bottlenecks, and structural dependency issues early in the design cycle. • Enhanced Collaboration: Standardizes technical documentation using global UML rules, helping developers, managers, and stakeholders communicate effectively. • Component Reusability: Allows teams to identify repeating patterns and design modular, reusable components across different projects.
Output Analysis	
Conclusion	I have successfully understood what is Software Engineering, SDLC, what is UML, types, UML components, Relationship, Symbols, and Relational Rose
Date	06/07/2026